

## Microsoft Helps Vatican Recreate St. Peter's Basilica Digitally

Technology company Microsoft is working with the Roman Catholic Church to create a detailed, digital version of St. Peter's Basilica in Vatican City.

The project aims to give people a new way to experience the famous building, which sits in the Catholic Church's spiritual and administrative center in Rome.

Roman Catholic leaders recently joined Microsoft's President Brad Smith to present the new digital creation to the public. The official launch of the system has been set for December 1. It will permit individuals to explore St. Peter's Basilica online and provide interactive experiences for all visitors to the Vatican.

The effort involved drones flying around the empty basilica for four weeks to capture images of all parts of the building. The drones – equipped with many cameras and lasers – took more than 400,000 high-quality pictures.

Images of St. Peter's Basilica are projected on a screen during a press exhibition 'Petros eni', which is part of the project "St. Peter's Basilica: AI-Enhanced Experience" at the Vatican, Saturday, Nov. 9, 2024. (AP Photo/Gregorio Borgia)

The process used artificial intelligence (AI) methods to create a complete 3D copy of the basilica. One goal of the project was to provide a way for anyone in the world to have the chance to “visit” the church and learn about its history.

Microsoft chief Smith told a press conference at Vatican City the project represented “one of the most technologically advanced and sophisticated projects of its kind” ever carried out.

Cardinal Mauro Gambetti is the Archpriest of St. Peter's Basilica. He praised the new technology at the press conference. Gambetti compared the highly detailed digital experience to looking up at a “starry sky on a summer night.” He added, “The new tools act like a telescope or spaceship for better viewing.”

Cardinal Mauro Gambetti, right, and Microsoft's vice-Chairman and President Brad Smith unveil the project "St. Peter's Basilica: AI-Enhanced Experience" during a press conference at the Vatican, Monday, Nov. 11, 2024. (AP Photo/Gregorio Borgia)

The project was launched ahead of the Vatican's 2025 Jubilee celebration. This marks a holy year at the Vatican when more than 30 million worshippers are expected to visit the basilica.

This will be in addition to the 50,000 people who visit on a normal day. The basilica will celebrate its 400th anniversary in 2026.

The Associated Press reported Pope Francis visited Smith and the project's development teams at the Vatican. He said the effort should help everyone "feel welcome in this great house."

The pope added that the project was a way to recognize and continue to celebrate the Vatican's rich spiritual history. "...It is a gift and a task to care for it, in both a spiritual and material sense, even through the latest technologies," he said.

Pope Francis has warned that AI should only be used in ethical ways. In his World Message of Peace this year, he proposed an international treaty to control, or regulate, the technology. He argued that AI systems lack human values and could cause great harm.

A photographer takes pictures during the preview to the press of the exhibition 'Petros eni', which is part of the project "St. Peter's Basilica: AI-Enhanced Experience" at the Vatican, Saturday, Nov. 9, 2024. (AP Photo/Gregorio Borgia)

The digital system permits visitors to request entry times to the basilica. This could be a welcome change for visitors used to waiting in lines for hours to get into the basilica, which is one of the world's most visited buildings.

The collected images have already identified structural damage and signs of deterioration. Experts said the technology can find such problems much faster and more effectively than human workers. Vatican officials plan to use the images to continually identify areas needing repairs.

Smith did not say how much Microsoft had invested in the project. But he said the idea came after Pope Francis urged technology companies in 2018 to come together to use AI for cultural purposes.

Smith noted his company had also completed similar AI projects at Mont Saint-Michel in France and for Ancient Olympia in Greece.

I'm Bryan Lynn.

The Associated Press, Reuters, Agence France-Presse and Vatican News reported on this story. Bryan Lynn adapted the reports for VOA Learning English.

## Vocabulary

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drone - คำนาม. อากาศยานไร้คนขับ

advanced - คำคุณศัพท์. ที่พัฒนาหรือก้าวหน้าไปสู่ขั้นสูง

sophisticated - คำคุณศัพท์. ที่พัฒนาไปสู่ระดับความซับซ้อนสูง

worshipper - คำนาม. ผู้สักการะ, ผู้บูชา

task - คำนาม. งาน, การกิจ

ethical - คำคุณศัพท์. ที่มีจริยธรรม, เป็นไปตามหลักจริยธรรม

deteriorate - คำกริยา. เสื่อมโทรม, ทรุดโทรมลง, แย่ลง

## Reading Comprehension Quiz

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**1. What is the primary goal of the collaboration between Microsoft and the Roman Catholic Church?**

- A) To construct a new physical basilica.
- B) To create a detailed, digital version of St. Peter's Basilica.
- C) To conduct extensive repairs on the existing basilica.
- D) To host an international technology conference at the Vatican.

**2. According to the article, what is the significance of St. Peter's Basilica's location in Rome?**

- A) It serves as a major tourist attraction only.
- B) It is primarily a historical museum for ancient artifacts.
- C) It functions as the Catholic Church's spiritual and administrative center.
- D) It is a meeting point for technology companies and religious leaders.

**3. When is the official launch of the digital system set, and what immediate functionality will it offer?**

- A) November 9, allowing press exhibitions only.
- B) December 1, permitting online exploration and interactive experiences.
- C) November 11, for a press conference announcement.
- D) December 25, to coincide with Christmas celebrations.

**4. How was the vast amount of imagery for the digital basilica collected?**

- A) By photographers using traditional cameras over several months.
- B) Through satellite imaging technology capturing aerial views.
- C) Using drones equipped with cameras and lasers for four weeks.
- D) By scanning existing architectural blueprints and historical documents.

**5. What advanced technology was pivotal in creating the complete 3D copy of the basilica, and what was one of its main objectives?**

- A) Virtual Reality (VR) to offer immersive games.
- B) Artificial Intelligence (AI) to enable global virtual visits and historical learning.

- C) Robotics for automated maintenance and cleaning.
- D) High-definition videography for live streaming events.

**6. How did Microsoft chief Brad Smith describe the St. Peter's Basilica digital project?**

- A) As a modest effort in cultural preservation.
- B) As a straightforward application of current technology.
- C) As one of the most technologically advanced and sophisticated projects of its kind.
- D) As an initial experiment with limited future potential.

**7. Cardinal Mauro Gambetti praised the new technology, comparing the digital experience to what?**

- A) Reading an intricate architectural guide.
- B) A simple walk through a museum.
- C) Looking up at a starry sky on a summer night, with tools like a telescope or spaceship.
- D) A standard documentary film about the Vatican.

**8. The digital project was launched in anticipation of what significant Vatican event, and what is the expected attendance?**

- A) The basilica's 400th anniversary in 2026, with fewer daily visitors.
- B) The 2025 Jubilee celebration, expecting more than 30 million worshippers.
- C) A normal tourist season, with approximately 50,000 visitors per day.
- D) A special exhibition showcasing new Vatican art, with limited access.

**9. What was Pope Francis's hope regarding the project after his visit to the development teams?**

- A) That it would increase the Vatican's financial income.
- B) That it would help everyone 'feel welcome in this great house.'
- C) That it would primarily attract technology enthusiasts.
- D) That it would reduce the need for physical pilgrimages.

**10. Beyond being welcoming, what broader purpose did Pope Francis highlight for the project?**

- A) To establish the Vatican as a leader in technological innovation.
- B) To replace traditional religious ceremonies with digital ones.
- C) To recognize and celebrate the Vatican's rich spiritual history, caring for it spiritually and materially through technology.
- D) To provide a model for other religious institutions to digitize their assets.

**11. What is Pope Francis's general concern regarding Artificial Intelligence, as stated in his World Message of Peace?**

- A) He believes AI should be adopted without any restrictions.
- B) He advocates for an international treaty to control AI, citing its lack of human values and potential for harm.
- C) He suggests AI should be exclusively used for cultural and historical preservation.
- D) He sees AI as a benevolent force that will solve global issues.

**12. How is the new digital system expected to benefit visitors planning a physical visit to the basilica?**

- A) It will offer free entry for all digital users.
- B) It allows visitors to request entry times, potentially reducing long queues.
- C) It provides virtual reality tours that eliminate the need for physical visits.
- D) It offers exclusive access to restricted areas not open to the general public.

**13. What unexpected but valuable insight has been gained from the collected images of the basilica?**

- A) They have revealed new artistic discoveries within the building.
- B) They have identified structural damage and signs of deterioration, aiding faster repairs.
- C) They have helped to trace the historical evolution of its construction.
- D) They have been used to create a new promotional video for the Vatican.

**14. According to Microsoft's Brad Smith, what was the impetus behind the development of this project?**

- A) Microsoft's desire to expand its market share in Italy.
- B) A direct request from the Italian government for cultural heritage preservation.
- C) Pope Francis's 2018 call for technology companies to use AI for cultural purposes.
- D) A competitive initiative among major tech companies to partner with religious institutions.

**15. In which other locations has Microsoft implemented similar AI-enhanced cultural projects?**

- A) Only within other major cities in Italy.
- B) At Mont Saint-Michel in France and for Ancient Olympia in Greece.
- C) Across various historical sites in the Middle East.
- D) Exclusively for modern architectural marvels worldwide.